Causal Rejected-Side Masking for Reasoning Preference Optimization

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Abstract

Direct Preference Optimization (DPO) penalizes rejected responses as whole trajectories. This is a poor inductive bias for multi-step reasoning: a rejected solution can contain a correct prefix, a localized first error, and later steps that merely inherit the wrong state. We study causal rejected-side masking, a family of DPO-style objectives that changes only how the rejected response is aggregated. The central operation is prefix protection: steps before the first error are masked out of the rejected-side penalty. Causal-Masked DPO (CM-DPO) further adds a decayed penalty on post-error suffix steps. Controlled arithmetic experiments show that vanilla DPO strongly degrades reasoning accuracy, while causal masking improves retention and held-out performance. Across three harder-arithmetic seeds, CM-DPO is best on average, but only narrowly over a first-error-only baseline. Additional diagnostics show that truncating rejected traces suppresses errors but can still hurt correct prefixes, and that off-by-one first-error labels can break the intended mask. GSM8K-style pilots are mixed: prefix protection helps, but suffix decay has not yet produced a reliable additional gain. We therefore present CM-DPO as a causal masking framework, with prefix protection as the robust empirical finding and suffix decay as a promising but localization-sensitive design choice.

1 Introduction

Preference optimization is widely used to adapt language models to desired behavior. Direct Preference Optimization (DPO) (Rafailov et al., 2023) is particularly attractive because it trains directly on preferred and rejected responses without fitting a separate reward model. However, the standard DPO objective treats a rejected response as uniformly negative evidence. This is a natural simplification for short answers, but a questionable one for long reasoning traces.

The problem is causal credit assignment. A rejected mathematical solution can correctly understand the question, set up the right equation, and then make one local arithmetic mistake. Penalizing the entire rejected trajectory suppresses both the actual error and the correct reasoning prefix. This creates an over-penalty failure mode: the model is trained away from useful intermediate reasoning because that reasoning happened to appear inside a rejected final answer.

We study a simple alternative: causal rejected-side masking. Given a rejected response segmented into steps and an estimate of the first erroneous step, we change only the rejected-side aggregation in DPO. Steps before the first error are protected from negative pressure. The first error receives full penalty. Later steps can either be fully penalized, ignored, or decayed. Causal-Masked DPO (CM-DPO) is one point in this family, using prefix protection plus exponential suffix decay. The framework keeps the same preference-pair interface as DPO and requires only step boundaries and a first-error estimate.

Our empirical results suggest a more nuanced story than the strongest version of the initial hypothesis. On controlled harder arithmetic data, causal masking clearly avoids the severe degradation caused by vanilla DPO. In the strongest multi-seed setting, CM-DPO achieves the best mean held-out accuracy. At the same time, its margin over first-error-only DPO is small. On GSM8K-style pilots, prefix protection remains useful, but suffix decay has not yet shown a reliable benefit.

We therefore frame the contribution conservatively. The robust empirical claim is not that one fixed suffix-decay schedule is universally best. Rather, rejected reasoning trajectories contain causal structure, and DPO benefits from respecting at least the most important part of that structure: correct prefixes should not be penalized as if they were errors.

Our contributions are:

- We formulate trajectory-level DPO over-penalty as a causal credit-assignment problem in multi-step reasoning.
- We introduce a rejected-side masking family that includes vanilla DPO, prefix-masked DPO, first-error-only DPO, and CM-DPO as different weight choices under one objective.
- We implement token-level masked DPO with LoRA training and first-error localization, enabling direct comparisons under the same model, data, and optimization setup.
- We report controlled arithmetic and GSM8K-style pilots showing that prefix protection is robustly useful, while suffix decay remains task- and data-quality-sensitive.
- We add diagnostic controls for suffix schedules, truncated rejected responses, and off-by-one localization noise, clarifying when the masking mechanism succeeds and when it fails.

2 Related Work

Preference optimization. Reinforcement learning from human feedback aligns language models by learning a reward model and optimizing against it (Ouyang et al., 2022). DPO (Rafailov et al., 2023) removes the explicit reward-model stage and directly optimizes preference pairs through policy-reference likelihood ratios. Our work keeps DPO's trajectory-level data interface, but modifies how rejected trajectories contribute to the objective. This makes the proposal orthogonal to the source of the preference pair: it can be applied to human preferences, verifier-generated preferences, or synthetic reasoning preferences when a first-error signal is available.

Reasoning supervision. Benchmarks such as GSM8K (Cobbe et al., 2021) and MATH (Hendrycks et al., 2021) emphasize multi-step mathematical reasoning. Process supervision and step-level verification aim to supervise intermediate reasoning rather than only final answers (Lightman et al., 2024). Causal rejected-side masking is complementary: it can consume first-error signals from verifiers or judges, but does not require transforming the data into explicit step-level preference pairs.

Credit assignment in reasoning traces. Outcome-level supervision gives a single label to an entire solution. This is convenient, but it conflates the step that caused failure with earlier steps that may be correct and later steps that may be consequences of the first error. Process supervision addresses this by labeling intermediate states. Our setting lies between these extremes: we assume a first-error localization signal, then use it to reweight the rejected side of a standard preference objective. The goal is not to replace process supervision, but to use limited process information inside a DPO-compatible loss.

Recent process-reward and first-wrong-step methods, including OmegaPRM, HRM, and Conditional Reward Modeling, pursue denser models of intermediate correctness. Those approaches can provide richer supervision than the single first-error index used here. The tradeoff is interface complexity: they typically require training or querying a separate process model. Causal masking is lighter-weight and can be layered on top of such systems by using their scores to localize the first error or to replace hard masks with calibrated soft weights.

Counterfactual and causal preference objectives such as GC2PO are also motivated by the need to separate causal reasoning steps from incidental tokens. Our objective is less ambitious: it does not enforce counterfactual invariances or learn a dense causal model of the trace. Instead, it tests whether a first-error signal is sufficient to repair a specific DPO failure mode, namely over-penalizing correct rejected prefixes.

Masked and weighted objectives. The method can be viewed as a token-weighted variant of DPO in which weights are induced by step-level causal structure. The preferred response remains fully weighted, while the rejected response receives structured weights. This differs from generic loss reweighting because the mask is asymmetric and causal: tokens before the first error in the rejected response are not treated as negative evidence. Token-level reweighting methods for long chain-of-thought supervised fine-tuning, such as VCORE-style objectives, make a related observation that not all reasoning tokens should carry equal training weight. CM-DPO applies the same broad principle to the rejected side of preference optimization rather than to positive-only supervised learning.

Parameter-efficient adaptation. Our experiments use LoRA (Hu et al., 2022) to make repeated preference-optimization comparisons feasible. The masking objective is independent of LoRA and can also be used with full fine-tuning.

3 Method

3.1 DPO and rejected-side over-penalty

For a prompt x, DPO uses a preferred response y_w and rejected response y_l. Its implicit reward is

$$r_\theta(y \mid x) = \log \pi_\theta(y \mid x) - \log \pi_{\text{ref}}(y \mid x), \tag{1}$$

and the loss is

$$\mathcal{L}_{\text{DPO}}(\theta) = -\mathbb{E}\left[\log \sigma\left(\beta\left(r_\theta(y_w \mid x) - r_\theta(y_l \mid x)\right)\right)\right]. \tag{2}$$

When y_l is a multi-step reasoning trace, this objective penalizes all rejected tokens even if some earlier steps are correct. The failure is specific to the rejected side: the preferred response should remain positive evidence, but a rejected response may contain both positive and negative reasoning evidence.

3.2 Causal step weights

Let the rejected response be segmented into steps

$$y_l = (s_1, \ldots, s_K), \tag{3}$$

and let m be the estimated first-error index. A causal rejected-side mask assigns each rejected step a nonnegative weight w_k. Steps with $k < m$ occurred before the localized error and should not receive negative pressure from the rejected label. The preferred response remains fully weighted. The weighted rejected reward is

$$r_\theta^w(y_l \mid x) = \sum_{k=1}^{K} w_k \left[\log \pi_\theta(s_k \mid x, s_{<k}) - \log \pi_{\text{ref}}(s_k \mid x, s_{<k})\right]. \tag{4}$$

The masked DPO loss is

$$\mathcal{L}_{w-\text{DPO}}(\theta) = -\mathbb{E}\left[\log \sigma\left(\beta\left(r_\theta(y_w \mid x) - r_\theta^w(y_l \mid x)\right)\right)\right]. \tag{5}$$

3.3 Variants

Table 1 shows the four weight patterns used in the experiments. These variants isolate two design choices: whether to protect the pre-error prefix, and how much penalty to assign after the first error.

CM-DPO therefore adds suffix decay to the robust prefix-protection idea. This distinction matters empirically: the prefix mask is consistently useful, while the extra decay design is not always better than simpler alternatives.

Table 1: Rejected-side causal masking family. The preferred response is always fully weighted.

Variant	Prefix $k < m$	First error $k = m$	Suffix $k > m$
Vanilla DPO	1	1	1
Prefix-masked DPO	0	1	1
First-error-only DPO	0	1	0
CM-DPO	0	1	γ^{k-m}

3.4 Interpretation and normalization

The objective is a DPO-style credit-assignment objective, not a claim that the one-sided weighted loss is exactly equivalent to canonical trajectory-level DPO under the usual Bradley–Terry reward interpretation. Standard DPO compares two complete trajectories with unweighted policy-reference log-ratios. In contrast, our objective changes only the rejected-side aggregation before forming the same logistic preference loss. This is deliberate: the method encodes the assumption that a failed reasoning trace contains heterogeneous evidence, with correct pre-error tokens, a localized causal error, and consequence tokens.

In the current implementation, weighted rejected log-likelihoods are summed rather than normalized by the total mask mass. This keeps the amount of rejected-side evidence proportional to the number and weight of penalized tokens, but it also means different masks can change the effective gradient scale. We therefore compare variants under matched optimizer settings and include likelihood probes that report span-specific changes. A normalized variant, where each example is divided by $\sum_t w_t$, is an important control for future larger-scale experiments because it separates mask structure from total penalty mass.

3.5 Implementation

Step-level weights are expanded to token-level response masks by aligning segmented step character spans to tokenizer offsets. Each response token receives the weight of the step whose character span overlaps that token; prompt tokens are excluded from the rejected-response mask. The DPO trainer computes policy and reference log-likelihoods with these masks applied to the rejected response. The reference model is frozen, and all reported model updates use LoRA adapters. If a rejected response cannot be segmented, cannot be localized, or has an ambiguous first-error index, the example is filtered rather than assigned a high-confidence causal mask.

3.6 Training procedure

The full procedure is intentionally close to standard DPO. For each preference pair, we segment the rejected response, localize the first erroneous step, choose a rejected-side weight schedule from Table 1, and expand the resulting step weights to token weights. Training then uses the same policy-reference likelihood ratio as DPO, with the weighted rejected reward replacing the unweighted rejected reward. This design isolates the causal credit-assignment intervention: the optimizer, reference model, preferred-response treatment, and LoRA adaptation setup are unchanged across variants.

4 Experiments

4.1 Controlled harder arithmetic

The main mechanism experiments use synthetic harder arithmetic preference pairs. Each example contains a prompt, a correct step-by-step solution, and a rejected solution with a localized arithmetic or state-transition error. This setting gives high-quality first-error labels and directly tests whether rejected-side masking protects correct prefixes.

Table 2: Harder arithmetic eval500 after harder2000 training. Mean and standard deviation are over three seeds.

Model	Accuracy	Std.
Base	0.410	–
Vanilla DPO	0.253	0.054
Prefix-masked DPO	0.365	0.012
First-error-only DPO	0.427	0.016
CM-DPO	0.433	0.011

We train Qwen2.5-0.5B-Instruct with LoRA adapters. The key scaled setting uses 2,000 preference pairs, $\gamma = 0.25$, one epoch, $\beta = 0.1$, maximum length 768, batch size 1, and gradient accumulation 8. We evaluate on 500 held-out harder arithmetic problems.

The controlled setting is intentionally narrow. It is designed to test the mechanism under clean localization rather than to claim broad mathematical ability. This makes failures informative: if vanilla DPO harms accuracy even when the preference data are clean, then the problem is not only noisy labels but also the way rejected reasoning is aggregated.

4.2 GSM8K-style pilots

We also run GSM8K-style pilots to test transfer beyond controlled templates. We generate preference pairs from GSM8K prompts using the base model, select sampled correct responses when available, and otherwise fall back to the gold answer. Rejected solutions are localized using an API judge that returns the first erroneous step and confidence. We filter examples by confidence, non-final first-error position, and truncation heuristics.

The GSM8K pilots are intentionally treated as diagnostic rather than final benchmark evidence. They reveal whether the same masking idea survives noisier generated reasoning, but they also expose data-quality issues such as gold-fallback chosen responses and style mismatch.

4.3 Evaluation questions

We evaluate the masking family through five questions. First, does vanilla DPO degrade reasoning behavior when rejected traces contain correct prefixes? Second, does masking the pre-error prefix recover performance? Third, does assigning a decayed penalty to post-error suffix steps improve over simpler first-error-only or prefix-masked alternatives? Fourth, can a truncated-rejected baseline explain the effect without explicit masks? Fifth, how sensitive is the method to first-error localization noise? The first two questions are the core causal-masking claims; the third is the specific additional hypothesis behind CM-DPO; the last two are robustness and alternative-explanation checks.

In addition to generation accuracy, we report deterministic likelihood probes on the rejected response. These probes measure the change in average masked log-likelihood for pre-error prefixes, first-error steps, post-error suffixes, and the full CM-DPO mask. They are useful because they directly test whether an adapter protects correct prefixes and suppresses localized errors, independent of decoding noise.

5 Results

5.1 Harder arithmetic supports causal masking

Table 2 shows the 500-example harder arithmetic evaluation after training on 2,000 preference pairs over three seeds. Vanilla DPO is consistently harmful. Prefix-masked and first-error-only DPO are much stronger. CM-DPO is best on average, but its margin over first-error-only DPO is small.

Table 3: Example rejected trace and CM-DPO weights with $\gamma = 0.25$. The first erroneous step is marked in bold.

Step	Rejected reasoning step	Weight
1	Adult tickets: $19 \times 14 = 266$ dollars.	0
2	Child tickets: $9 \times 6 = 63$ dollars.	1
3	Subtotal: $266 + 63 = 329$ dollars.	0.25
4	Final total: $329 - 20 + 29 = 338$.	0.0625
5	Final answer: 338.	0.0156

This result supports the central causal-masking claim: uniformly penalizing rejected reasoning is unsafe, and protecting pre-error reasoning substantially improves behavior. It does not, by itself, prove that suffix decay is much better than simply training on the first erroneous step.

5.2 Ablation: prefix protection versus suffix decay

The four variants decompose the effect of causal masking. Vanilla DPO has no prefix protection and drops from a base accuracy of 0.410 to 0.253. Prefix-masked DPO changes only the pre-error prefix weights and recovers much of the lost performance, reaching 0.365. First-error-only DPO removes suffix pressure entirely and reaches 0.427. CM-DPO keeps a decayed suffix penalty and reaches 0.433.

This pattern supports a clear primary conclusion and a cautious secondary one. The primary conclusion is that prefix protection is responsible for the largest improvement over vanilla DPO. The secondary conclusion is that suffix decay can help in the controlled setting, but the observed gain over first-error-only DPO is small relative to the current scale of the experiments.

5.3 Qualitative mask example

Table 3 shows a representative harder-arithmetic rejected trace. The first step is correct, but the second step makes a local arithmetic error: 9×6 is written as 63 rather than 54. Later steps then inherit the wrong subtotal. Vanilla DPO penalizes the entire rejected response, including the correct adult-ticket computation. CM-DPO masks that prefix, applies full penalty to the first erroneous step, and assigns decayed weights to the inherited suffix.

This example illustrates the intended causal credit assignment. The rejected final answer is wrong, but not every token in the rejected trace is equally wrong. The correct prefix is useful evidence about the task, the first error is the main negative evidence, and the suffix is a consequence of the erroneous state.

5.4 Suffix schedule controls the error–suffix tradeoff

Table 4 reports a deterministic likelihood probe after harder2000 training. The endpoints correspond to first-error-only DPO ($\gamma = 0$) and prefix-masked DPO ($\gamma = 1$). All settings strongly increase the likelihood of pre-error prefixes, confirming that causal masks protect correct rejected prefixes. Increasing γ makes the first-error penalty stronger, but it also increasingly penalizes post-error suffix tokens. The $\gamma = 0.75$ setting gives the strongest first-error suppression among decayed schedules, while $\gamma = 1$ behaves like full suffix penalty and heavily suppresses the suffix.

This ablation clarifies why suffix decay should be treated as a schedule rather than a single fixed claim. Small and moderate γ values preserve prefix likelihood and suppress the first error without aggressively suppressing inherited suffix text. Large γ values more closely resemble prefix-masked DPO: they apply strong pressure to all post-error tokens, which can be useful for error suppression but risks treating consequence steps as independently wrong.

Table 4: Harder2000 likelihood probe for CM-DPO suffix schedules. Entries are post-training minus pre-training masked log-likelihood changes on rejected spans. Higher prefix delta means better prefix retention; more negative error delta means stronger first-error suppression.

Schedule	Prefix Δ	Error Δ	Suffix Δ	Mask Δ
$\gamma = 0$	35.112	-18.082	17.304	-15.091
$\gamma = 0.25$	35.234	-20.470	16.688	-17.658
$\gamma = 0.5$	35.320	-20.866	15.317	-18.392
$\gamma = 0.75$	35.254	-23.838	11.550	-22.246
$\gamma = 1$	35.102	-27.394	-44.424	-35.863

Table 5: Harder2000 reviewer diagnostics. Entries are post-training minus pre-training masked log-likelihood changes on the original rejected spans. Higher prefix delta means better prefix retention; more negative error delta means stronger suppression of the true first error.

Variant	Prefix Δ	Error Δ	Suffix Δ	Mask Δ
Truncated rejected	-3.938	-40.439	17.413	-37.157
Clean CM-DPO	35.218	-20.436	16.718	-17.608
First-error shifted +1	35.700	12.423	5.526	12.492

5.5 Reviewer baselines and localization robustness

Table 5 adds two diagnostics motivated by the main threats to validity. First, a truncated-rejected baseline keeps only the rejected trace through the first error and then trains standard DPO on that shorter rejected response. This baseline strongly suppresses the first-error span, but it also reduces the likelihood of the correct pre-error prefix. Thus, the gain from causal masking is not merely caused by shortening rejected responses; explicit prefix protection changes which evidence is treated as negative.

Second, we inject a controlled localization error by shifting the first-error index one step later before constructing the CM-DPO mask. This off-by-one shift protects the true first error and penalizes later consequence steps. The resulting adapter no longer suppresses the true error span. This confirms that first-error quality is a real dependency of the method, and motivates confidence filtering or soft masks when localization is uncertain.

5.6 GSM8K pilots are mixed

Table 6 summarizes the GSM8K-style pilots on a 100-example evaluation set. As localization and filtering improve, all trained variants can beat the base model. However, CM-DPO does not dominate. Prefix-masked DPO is best in the 109-example high-quality setting, while first-error-only is best in the more-candidates setting.

5.7 Localization and data quality

Table 7 shows why the GSM8K pilots should be interpreted cautiously. API judge localization greatly improves the fraction of examples with non-final first errors, but strict high-quality filtering still leaves limited data. Increasing candidates raises the sampled-correct chosen rate in raw data, yet does not improve downstream accuracy after filtering.

The robust conclusion is that prefix protection matters. The suffix-decay component of CM-DPO remains plausible on controlled arithmetic, but the GSM8K pilots do not yet establish it as a reliable additional improvement.

Table 6: GSM8K-style eval100. Entries are correct out of 100.

Training data	Base	Vanilla	Prefix	First-error	CM-DPO
Cheap localizer, 49 rows	16	19	22	21	20
API judge, 49 rows	16	18	25	23	20
API judge HQ, 109 rows	16	25	29	28	28
More-candidates HQ, 87 rows	16	22	25	28	27

Table 7: GSM8K-style localization and filtering diagnostics.

Dataset	Rows	Non-final first error	Sampled chosen
Cheap localizer	49	1 / 49	26 / 49
API judge	49	26 / 49	26 / 49
API judge raw	199	121 / 199	77 / 199
API judge HQ	109	109 / 109	34 / 109
More-candidates raw	200	101 / 200	101 / 200
More-candidates HQ	87	87 / 87	35 / 87

6 Discussion

The experiments suggest separating the method into two claims. The first claim is prefix protection: rejected reasoning trajectories should not be uniformly penalized when their prefixes are correct. This claim is supported across controlled arithmetic and GSM8K-style pilots. The second claim is suffix decay: after the first error, later steps should receive a decayed but nonzero penalty. This claim is supported by some controlled likelihood evidence, but not yet by the GSM8K pilots.

This distinction changes the paper's center of gravity. Instead of presenting CM-DPO as a fixed objective whose suffix schedule is always best, we present causal rejected-side masking as a framework. CM-DPO is one instance of the framework. Prefix-masked and first-error-only objectives are not merely weak baselines; they are important ablations that reveal which part of the causal mask is doing work.

For future experiments, the most useful direction is not simply more GSM8K data. The data must contain style-matched chosen and rejected responses, high-confidence first-error labels, and enough post-error suffix content for suffix decay to matter. Without those properties, GSM8K training mostly tests task adaptation and prefix protection, not the full CM-DPO hypothesis.

This framing also affects how negative results should be interpreted. If CM-DPO does not beat first-error-only DPO on a noisy dataset, that does not necessarily refute causal masking. It may indicate that the available suffix tokens are low-value training signal, that localization is too noisy, or that the model benefits most from correcting the first error directly. The framework makes these possibilities experimentally separable.

The suffix-schedule ablation reinforces this interpretation. Larger γ values produce stronger first-error suppression, but they also move the objective toward full post-error penalty. This creates a tunable tradeoff between correcting the localized error and avoiding over-penalty on inherited suffix reasoning. The current results support using γ as a controlled hyperparameter rather than treating a single decay value as universally optimal.

The reviewer diagnostics sharpen two boundary conditions. Truncating the rejected response at the first error is a useful baseline because it avoids suffix penalties without requiring token masks. However, our likelihood probe shows that truncation can still reduce the likelihood of correct pre-error reasoning, whereas clean causal masks preserve it. Conversely, an off-by-one first-error shift can remove pressure from the true erroneous step. This means causal masking is only as good as its localization signal; in realistic settings, hard masks should likely be combined with confidence thresholds, soft protection, or PRM-style uncertainty estimates.

Finally, the objective should be understood as an asymmetric reweighting intervention rather than a complete replacement for reward modeling. If a strong process reward model can score intermediate steps directly, it may provide better first-error labels or continuous token weights. The contribution here is narrower: when the training interface is still pairwise DPO, a small amount of causal structure can prevent the rejected label from treating correct intermediate reasoning as negative evidence.

7 Limitations

The current evidence is preliminary. The strongest positive results come from controlled harder arithmetic, where first-error labels are cleaner than in open-ended model generations. This makes the mechanism easy to test but does not prove broad reasoning generalization.

The GSM8K pilots are intentionally reported as mixed evidence. They show that the pipeline runs end-to-end and that prefix protection helps, but they do not show a reliable suffix-decay benefit. Many high-quality filtered examples still use gold-fallback chosen responses, creating a style mismatch with model-generated rejected responses.

The current experiments also use Qwen2.5-0.5B-Instruct with LoRA. Larger models, stronger sampled solution generators, and better first-error judges may change the balance between prefix-masked, first-error-only, and CM-DPO objectives.

The method assumes usable step segmentation and first-error localization. The controlled off-by-one experiment shows that a one-step shift can protect the true error and therefore reverse the intended training signal. This is the main practical risk of hard causal masks. A production version should use calibrated localization confidence, soft masks, or verifier-derived uncertainty instead of treating every predicted first-error index as exact.

The current loss also uses unnormalized summed token weights on the rejected side. This is simple and preserves the intended amount of negative evidence, but it does not isolate causal structure from total gradient scale. Larger studies should include per-example weight normalization and uniform down-weighting baselines.

Generation accuracy is also sensitive to decoding configuration and prompt formatting, especially for small instruction-tuned models. We therefore use likelihood probes as mechanism evidence and treat generation accuracy as a downstream behavioral check rather than the only metric.

Finally, the paper does not yet include standard full-scale GSM8K or MATH evaluations. Those should be added only after the preference data construction yields enough style-matched sampled-correct chosen responses. The current draft should therefore be read as a mechanism paper in progress, not as a complete benchmark claim.

8 Conclusion

We studied causal rejected-side masking for reasoning preference optimization. The central empirical finding is that DPO should not treat partially correct rejected reasoning as uniformly negative: protecting correct prefixes consistently improves behavior relative to vanilla DPO. CM-DPO extends this idea with suffix decay and performs best on average in controlled harder arithmetic, but GSM8K-style pilots show that suffix decay is not yet a robust advantage over simpler masks. Future work should focus on cleaner style-matched preference data and stronger first-error localization before claiming broad GSM8K gains.

References

Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher

Hesse, and John Schulman. Training verifiers to solve math word problems. In arXiv preprint arXiv:2110.14168, 2021.

Dan Hendrycks, Collin Burns, Saurav Kadavath, Akul Arora, Steven Basart, Eric Tang, Dawn Song, and Jacob Steinhardt. Measuring mathematical problem solving with the MATH dataset. In Thirty-fifth Conference on Neural Information Processing Systems Datasets and Benchmarks Track, 2021.

Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In International Conference on Learning Representations, 2022.

Hunter Lightman, Vineet Kosaraju, Yuri Burda, Harrison Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. Let's verify step by step. In International Conference on Learning Representations, 2024.

Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll L. Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, John Schulman, Jacob Hilton, Fraser Kelton, Luke Miller, Maddie Simens, Amanda Askell, Peter Welinder, Paul Christiano, Jan Leike, and Ryan Lowe. Training language models to follow instructions with human feedback. In Advances in Neural Information Processing Systems, 2022.

Rafael Rafailov, Archit Sharma, Eric Mitchell, Christopher D. Manning, Stefano Ermon, and Chelsea Finn. Direct preference optimization: Your language model is secretly a reward model. In Advances in Neural Information Processing Systems, 2023.